EXPERIMENT - 2

AIM - To convert the ER diagram of an Indian e-commerce platform into a relational schema and implement the schema using My SQL CREATE TABLE statements with primary keys, foreign keys, NOT NULL, UNIQUE, ON DELETE CASCADE and ON DELETE SET NULL constraints. To insert sample data and demonstrate referential integrity violations.

SOURCE CODE-

```
CREATE DATABASE IndianECommerce;
USE IndianECommerce;

--1 CUSTOMER TABLE

CREATE TABLE Customer (
    Customer_ID INT PRIMARY KEY,
    First_Name VARCHAR (30) NOT NULL,
    Middle_Name VARCHAR (30),
    Last_Name VARCHAR (30) NOT NULL,
    Email VARCHAR (50) NOT NULL UNIQUE
);

--2 ADDRESS TABLE

CREATE TABLE Address (
    Address_ID INT PRIMARY KEY,
    House_No VARCHAR (20) NOT NULL,
    Street VARCHAR (50) NOT NULL,
    City VARCHAR (30) NOT NULL,
    State VARCHAR (30) NOT NULL,
```

```
    PIN_Code VARCHAR (10) NOT NULL,
    Customer_ID INT NOT NULL,
    FOREIGN KEY (Customer_ID)
        REFERENCES Customer (Customer_ID)
        ON DELETE CASCADE
);

--3 SELLER TABLE
CREATE TABLE Seller (
    Seller_ID INT PRIMAR KEY,
    Seller_Name VARCHAR (50) NOT NULL,
    Contact_No VARCHAR (15) NOT NULL UNIQUE
);

--4 CATEGORY TABLE
CREATE TABLE Category (
    Category_ID INT PRIMARY KEY,
    Category_Name VARCHAR (50) NOT NULL UNIQUE
);

--5 PRODUCT TABLE
CREATE TABLE Product (
    Product_ID INT PRIMARY KEY,
    Product_Name VARCHAR (100) NOT NULL,
    Price DECIMAL (10,2) NOT NULL,
    Stock INT NOT NULL,
    Category_ID INT NOT NULL,
    FOREIGN KEY (Category_ID)
        REFERENCES Category (Category_ID)
        ON DELETE CASCADE
);
```

```
--6  PRODUCT - SELLER TABLE

CREATE TABLE Product - Seller (
    Seller - ID  INT,
    Product - ID  INT,
    PRIMARY ( Seller - ID , Product - ID),
    FOREIGN ( Seller - ID)
        REFERENCES Seller ( Seller - ID)
        ON DELETE CASCADE,
    FOREIGN KEY (Product - ID)
        REFERENCES Product (Product - ID)
        ON DELETE CASCADE
);

--7  PHYSICAL PRODUCT TABLE

CREATE TABLE Physical - Product (
    Product - ID INT PRIMARY KEY,
    Weight DECIMAL (8,2),
    Dimensions VARCHAR (50),
    FOREIGN KEY (Product - ID)
        REFERENCES Product (Product - ID)
        ON DELETE CASCADE
);

--8  DIGITAL PRODUCT TABLE

CREATE TABLE Digital - Product (
    Product - ID INT PRIMARY KEY,
    File - size DECIMAL (10,2),
    Download - Link VARCHAR (255) UNIQUE,
    FOREIGN cKEY ( Product - ID)
        REFERENCES Product ( Product - ID)
        ON DELETE CASCADE
);
```

```
--9 ORDERS TABLE
CREATE TABLE Orders(
    Order_ID   INT  PRIMARY KEY,
    Order_Date DATE  NOT NULL,
    Customer_ID  INT NOT NULL,
    FOREIGN KEY (Customer_ID)
        REFERENCES Customer (Customer_ID)
        ON DELETE CASCADE
);

--10 ORDER ITEM TABLE
CREATE TABLE OrderItem (
    Order_ID  INT,
    Product_ID  INT,
    Quantity INT NOT NULL,
    Subtotal DECIMAL (10,2) NOT
    PRIMARY KEY (Order_ID, Product_ID),
    FOREIGN KEY (Order_ID)
        REFERENCES Orders(Order_ID)
        ON DELETE CASCADE,
    FOREIGN KEY (Product_ID)
        REFERENCES Product (Product_ID)
        ON DELETE CASCADE

);
--11 PAYMENT TABLE
CREATE TABLE Payment (
Payment_ID  INT PRIMARY KEY,
Payment_Date DATE DATE NOT NULL,
```

```
    Amount DECIMAL (10,2) NOT NULL,
    Payment_Mode VARCHAR (30) NOT NULL,
    Order_ID INT NOT NULL UNIQUE,
    FOREIGN KEY (Order_ID)
        REFERENCES Orders ( Order_ID )
        ON DELETE CASCADE
);

--12 DELIVERY TABLE

CREATE TABLE Delivery (
    Delivery_ID INT PRIMARY KEY,
    Delivery_Date DATE,
    Delivery_Status VARCHAR(30) NOT NULL,
    Order_ID INT NOT NULL UNIQUE,
    Address_ID INT,
    FOREIGN KEY (Order_ID)
        REFERENCES Orders (Order_ID)
        ON DELETE CASCADE,
    FOREIGN KEY (Address_ID)
        REFERENCES Address (Address_ID)
        ON DELETE SET NULL
);
```

```
-- INSERT SAMPLE DATA

-- CUSTOMER DATA

INSERT INTO Customer
VALUES
(1, 'Aarav', NULL, 'Sharma', 'aarav@gmail.com'),
(2, 'Anarya', 'K.', 'Verma', 'anarya@gmail.com'),
(3, 'Riya', NULL, 'Singh', 'riya@gmail.com');

-- ADDRESS DATA
INSERT INTO ADDRESS
VALUES
(101, '12A', 'Hazratganj', 'Lucknow', 'UP', '226001', 1),
(102, '45B', 'Gomti Nagar', 'Lucknow', 'UP', '226010', 2),
(103, '78C', 'Civil Lines', 'Prayagraj', 'UP', '211001', 3);

-- SELLER DATA
INSERT INTO Seller
VALUES
(201, 'Tech World', '9876543210'),
(202, 'Fashion Hub', '9876543211'),
(203, 'Digital Store', '9876543212');

-- CATEGORY DATA
INSERT INTO Category
VALUES
(301, 'Electronics'),
(302, 'Fashion'),
(303, 'Books');
```

```
-- PRODUCT DATA
INSERT INTO Product
VALUES
(401, 'Wireless Headphones', 1999.00, 50, 301),
(402, 'Smart Watch', 2999.00, 30, 301),
(403, 'Cotton T-shirt', 799.00, 100, 302),
(404, 'E-book Basics', 499.00, 999, 303);

-- PRODUCT-SELLER DATA
INSERT INTO Product-Seller
VALUES
(201, 401),
(201, 402),
(202, 403),
(203, 404);

-- PHYSICAL PRODUCT DATA
INSERT INTO Physical-Product
VALUES
(401, 0.35, '20 x 15 x 8 cm'),
(402, 0.25, '10 x 5 x 2 cm'),
(403, 0.20, '30 x 25 x 2 cm');

-- DIGITAL PRODUCT DATA
INSERT INTO Digital-Product
VALUES
(404, 25.50, 'https://example.com/python-basics');

-- ORDERS DATA
INSERT INTO Orders
VALUES
(501, '2026-08-20', 1),
(502, '2026-08-21', 2),
(503, '2026-08-22', 3);
```

```
--ORDER ITEM DATA
INSERT INTO Order Item
VALUES
(501, 401, 1, 1999.00),
(501, 403, 2, 1598.00),
(502, 402, 1, 2999.00),
(503, 404, 1, 499.00);
--PAYMENT DATA
INSERT INTO Payment
VALUES
(601, '2026-08-20', 3597.00, 'UPI', 501),
(602, '2026-08-21', 2999.00, 'Credit Card', 502),
(603, '2026-08-22', 499.00, 'UPI', 503);
--DELIVERY DATA
INSERT INTO Delivery
VALUES
(701, '2026-08-23', 'Delivered', 501, 101),
(702, '2026-08-24', 'Out for Delivery', 502, 102),
(703, '2026-08-25', 'Processing', 503, 103);
--DISPLAY TABLES
SELECT * FROM Customer;
SELECT * FROM Product;
SELECT * FROM Orders;
SELECT * FROM orderItem;
SELECT * FROM Payment;
SELECT * FROM Delivery;
```

```
--REFERENTIAL INTEGRITY VIOLATION 1
--Customer_ID 999 does not exist in Customer table

INSERT INTO Address
VALUES
(104,'10D','MG Road','Lucknow','Uttar Pradesh',
'226001', 999);

--REFERENTIAL INTEGRITY VIOLATION 2
--Product_ID 999 does not exist in Product table
INSERT INTO order_item
VALUES
(501, 999, 1, 1000.00);

--REFERENTIAL INTEGRITY VIOLATION 3
--Category_ID 999 does not exist in Category table
INSERT INTO Product
VALUES
(405, 'Invalid Product', 1000.00, 10, 999);

--REFERENTIAL INTGRITY VIOLATION 4
--Order_ID 999 does not exist in orders table

INSERT INTO Payment
values
(604, '2026-08-25', 1000.00, 'UPI', 999);

-DEMONSTRATION ON DELETE CASCADE AND DELETE SET NULL
DELETE FROM Product
WHERE Product_ID = 401;
DELETE FROM ADDRESS
WHERE ADDRESS_ID = 102;

SELECT * FROM Delivery;
```

OUTPUT- After executing the valid CREATE TABLE and INSERT statements, the tables are created successfully and the sample records are inserted.

The referential integrity violation statements produce errors because the inserted foreign-key values do not exist in their respective parent table. The ON DELETE CASCADE constraint automatically deletes related records when the referenced parent record is deleted, while ON DELETE SET NULL sets the corresponding foreign-key value to NULL when the referenced address is deleted.

